

Akwasi Poku Ofori Agyei

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Education

Kwame Nkrumah University of Science and Technology

Bachelor of Science in Computer Science | CWA: 79.52/100 (First Class)

Kumasi, Ghana

Jan 2023 – Sep 2026

Experience

AWS re/Start Trainee

Jan 2026 – May 2026

AmaliTech

Remote

- Completed intensive hands-on cloud training covering core AWS services including **Amazon EC2, Amazon S3, IAM, VPC, Amazon RDS, CloudWatch, and the AWS CLI**, with practical exercises in compute, object storage, identity and access management, networking, databases, and cloud monitoring.
- Practiced provisioning and managing Linux-based cloud environments, connecting to remote EC2 instances using SSH, configuring basic network and security settings, and using command-line tools to interact with AWS resources.

Software Engineer Intern

Sep 2025 – Nov 2025

Npontu Technologies

Tse Addo, Accra

- Evaluated an in-house language model against **GPT-5, Gemini, and LLaMA 3** using structured prompting and qualitative analysis to identify performance gaps and cultural biases in African-context responses.
- Contributed to a scalable full-stack application supporting thousands of concurrent user interactions by implementing authentication, payment processing, and database migrations using **PHP, Laravel, and Vue**.
- Improved mobile application responsiveness through component-based frontend architecture and structured state management using **React Native**.

Information Technology Support Intern

Sep 2024 – Nov 2024

Gigcorp Limited

Ridge, Accra

- Contributed in troubleshooting the Ghana Police eServices platform to identify system failures and validate expected system behaviour by stress testing the application and using log-driven investigation.
- Handled the reviewing and routing of online requests for police reports required for employment and travel purposes to the Criminal Data Services Bureau (CDSB) for processing and printing, and delivery.

Information Technology Support Intern

Oct 2023 – Dec 2023

Gigcorp Limited

Ridge, Accra

- Cleaned, validated, and preprocessed field-collected data used by proprietary geolocation software.
- Assisted with general office operations and day-to-day administrative tasks such as data entry, supporting smooth coordination of routine work in the office.

Projects

EcoWatch | Geospatial Machine Learning System | [ML Model](#) | [GitHub](#) | [Demo](#)

Aug 2026

- Independently developed the complete machine learning pipeline for detecting artisanal and small-scale gold mining regions from multispectral satellite imagery in Ghana, including data cleaning, preprocessing, partitioning, training, evaluation, and inference.
- Developed and evaluated a **13-band U-Net** semantic segmentation model, achieving **70.56% Dice** and **64.34% IoU** on a held-out test set of **427 satellite images**.
- Built an experimental training workflow using BCE–Dice loss, AdamW optimization, mixed-precision training, gradient clipping, and validation-based checkpoint selection, improving validation Dice from **47.46% to 74.50%**.

KNUST Chess Club Website | [Live Site](#)

Jan 2026

- Developed reusable frontend components using **React.js** to improve maintainability and expand the platform's capabilities for **200+** club members and visitors.

SmartCards | AI-Powered Learning Application | [GitHub](#) | [Live Demo](#)

Jun 2025

- Independently designed and deployed a full-stack AI application that transforms uploaded study materials into flashcards and multiple-choice questions using **gpt-3.5-turbo**.
- Developed a document ingestion and preprocessing pipeline supporting PDF, TXT, DOCX, and PPTX files, including text extraction, cleaning, and input-length management.
- Implemented a **Flask REST API**, **React** frontend, and Supabase authentication to support document processing and AI-assisted content generation.

Technical Skills

Languages: Python, JavaScript, C, C++, Java, PHP, SQL

ML/Data: Pandas, NumPy, PyTorch, Keras, JAX

Web Development: React, Flask, Next.js, Express, PHP, Laravel, Vue

Cloud/Tools: AWS, EC2, S3, IAM, VPC, RDS, CloudWatch, AWS CLI, Linux, Git, SSH

Certifications

AWS Certified Cloud Practitioner

Google DeepMind – AI Research Fundamentals – Four-course series covering small language models, language data, neural networks, and transformer architectures.